STHCIUX - Human-Computer Interactions and User Experience

MCO2 - Design Portfolio

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GOOGLE DRIVE - https://bit.ly/STHCIUX-GroceryApp
MIRO BOARD - https://miro.com/app/board/uXjVNUzMvbk=/
FIGMA PROTOTYPE - https://bit.ly/STHCIUX-GroceryApp-Figma

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Checkpoint 1

Brainstorming

The group started the meeting with the main goal of finding problems that could be solved using solutions.

As work models and personas were no longer required, we shifted our focus to the needfinding report as the main source where the group could extract pain points from the responses of the interviewees and survey respondents. All observed pain points were identified and compiled in Miro (a collaborative whiteboard program), which were then voted on by the members.

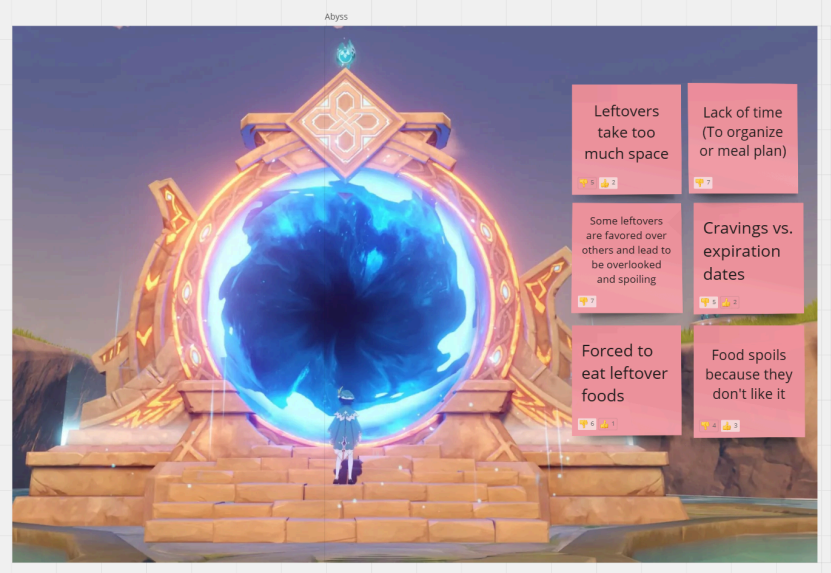


Figure 1. The space allocated for the pain points that were discarded

After the vote, the group retained the pain points with the most support and discarded those with less popular ones by way of voting.

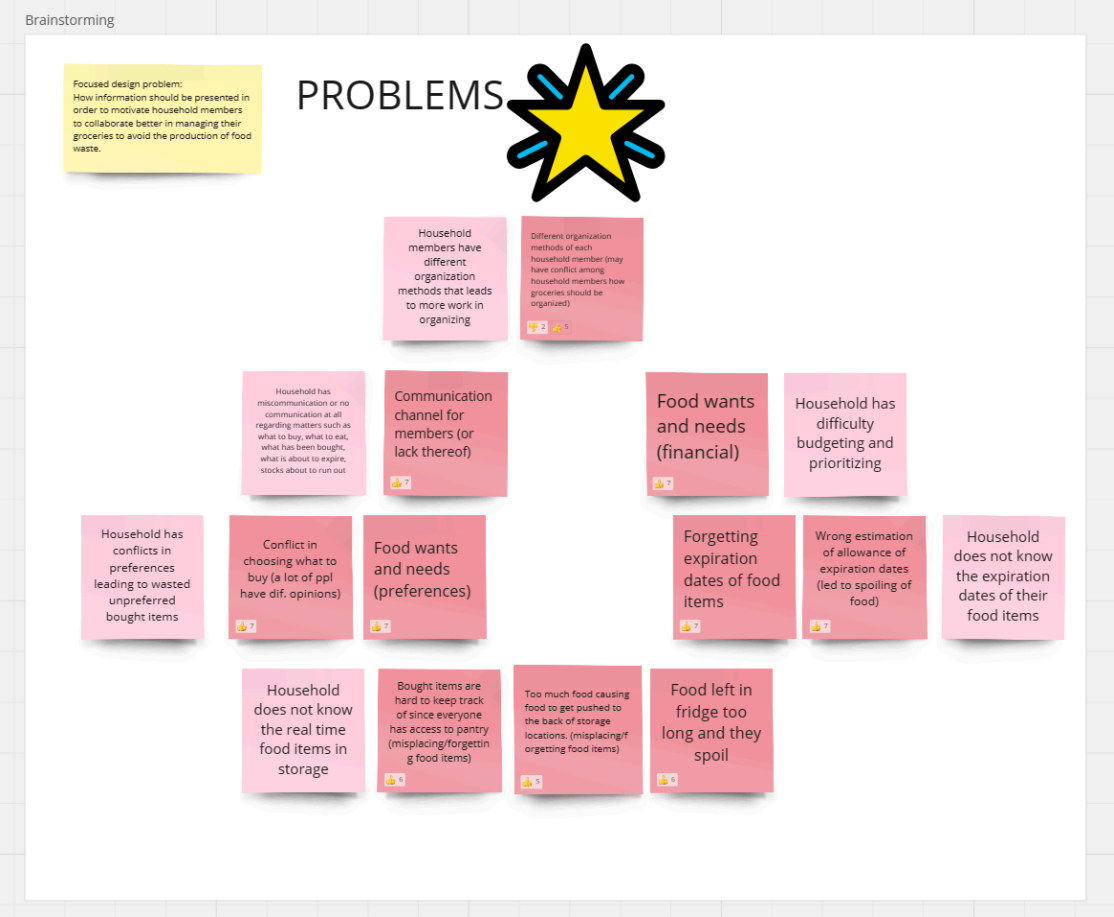


Figure 2. The retained pain points (in red) and the formulated problems (in pink)

The members then discussed which pain points were similar enough that they could be merged together to form a broader pain point. The problems were then properly defined based on the final pain points, where there were a total of 6 problems.

Before we delegated the problems to the different members of the group, we first voted between doing parallel design iterations and serial design iterations. The majority of the group voted for parallel design iterations, citing easier and more effective focus on specific problems. The group also voted between doing paragraph-based prototyping and storyboard-based prototyping, with a majority of the group voting for storyboard-based prototyping due to it being an easier way of preparing and explaining prototypes visually.

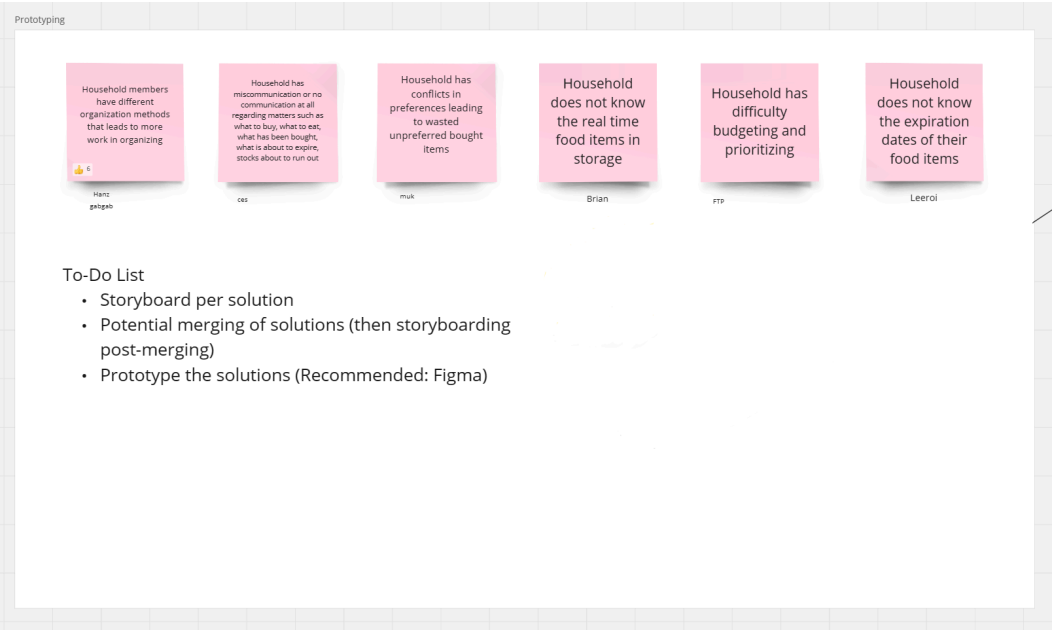


Figure 3. The problems and the member assignment

With six identified problems and seven team members, one problem needed to be tackled by two individuals. A vote was initiated to decide which problem was to be given to 2 members, with the general basis around the vote being the problem that seemed to be the most difficult. Afterward, the group members chose the problems they wanted to work on and create a prototype for.

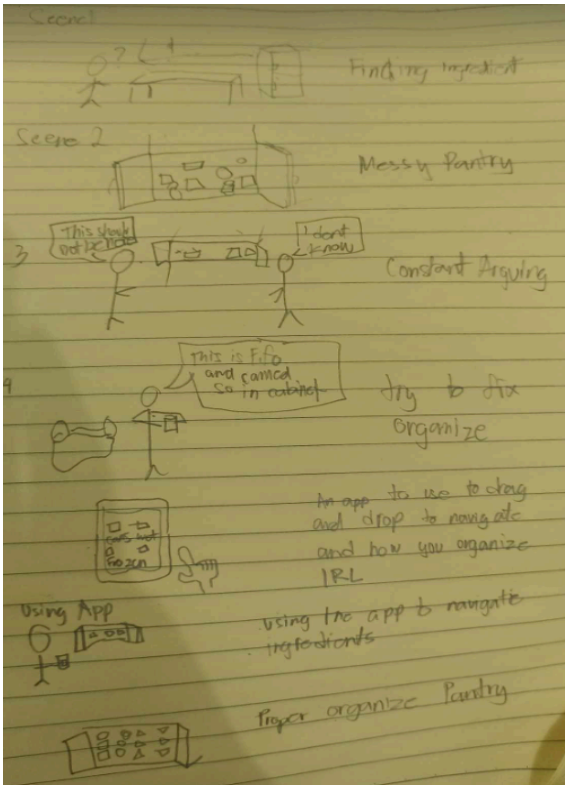
To conclude, the group came up with a simple to-do list for what should be done and achieved for checkpoint 1 of MCO 2. The group also scheduled meetings around the to-do list to ensure we remained aligned with our objectives.

Storyboard 1 - Problem 1

In this problem, it was decided that two members would work on the problem. This led to two designed solutions that were eventually combined.

Problem: Household members have different organization methods that lead to more work in organizing

Solution 1:



*Read from top to bottom

Narrative

I created this storyboard based on my experience of how we do things. As such, I thought of an organization app wherein all household members have access to the app. The app has folders/placeholders on what pantry or what part of their dedicated organization space is and is descriptive on each folder and holds on what it is there so that finding and organizing food items and groceries are made easier

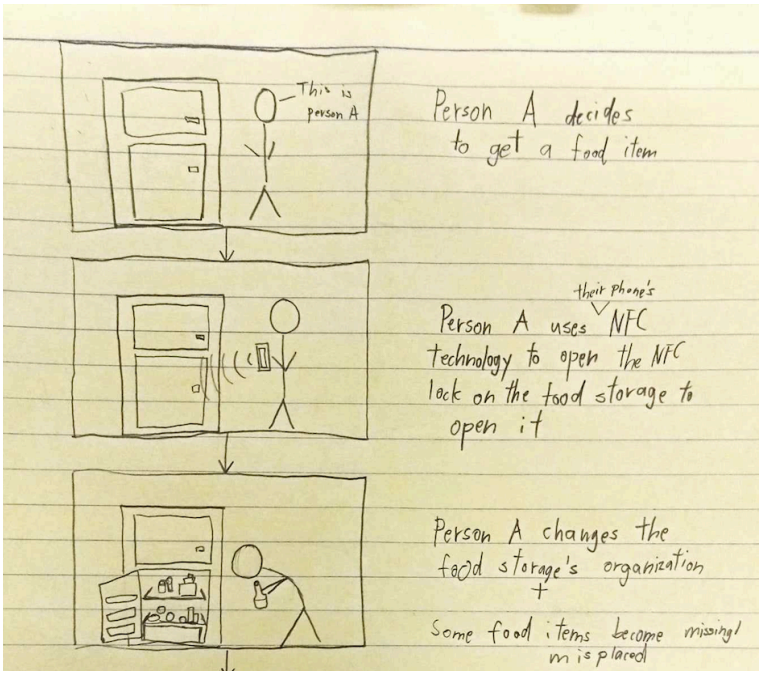
Evaluation

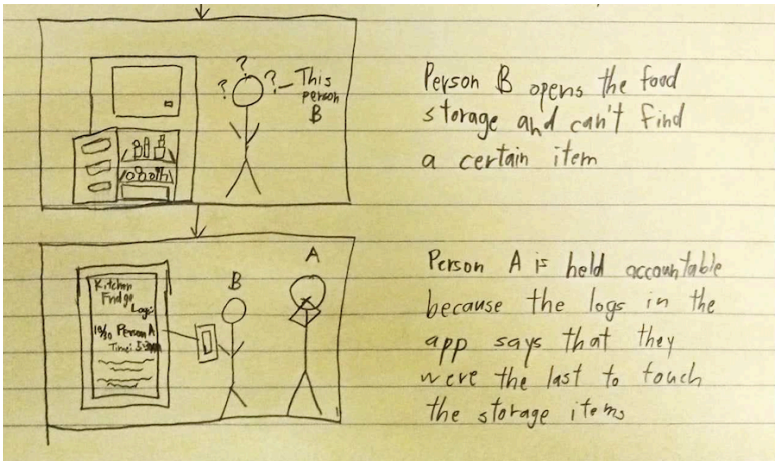
The evaluation of this solution was that it forced household members to organize their groceries twice: Once in real-life, and twice on the app. This led to a doubling of the household's work in terms of organizing their food items at home.

Key Insights

The main insight I have is the comment about the doubling of the work of household members makes good sense, and I have to find a way to avoid this mistake in the next iteration of my solution, preferably something similar to a system that automatically recognizes any item put in a food storage area through AI recognition.

Solution 2:





The image should read from top to bottom and left to right. The 5 part image shows the flow in which the proposed app/devices would be used in a practical setting

Narrative

When formulating a solution to the problem, I immediately thought of an accountability system. When people figure out who causes the disorganization or the reason for misplacing items, people will learn to be more mindful of how they interact with their food storage. I had to figure out a way that could help identify the key actors of the household who organize and use these storages. I thought of using QR code scanning for an app that would allow them to open food storages and this action would be logged on the app when they opened it, but I realized this is not as seamless to open and it would be really annoying to bring out the app just to open a food storage. That's when I recalled that there is an NFC technology embedded in modern smartphones that allows them to reload LRT Beep Cards. I figured that we could probably utilize this technology to seamlessly just tap or put our phones near the storage device lock and it would identify and open the food storage.

Evaluation

My groupmates felt that my solution in targeting accountability for this problem was a good idea. The use of NFC technology was seen to be feasible, and the user logging to help family members be more mindful was simple enough. The main concern about this from my peers was that it had ethical or cultural issues with it. In Filipino culture, a culture of sharing or “no damutan” is encouraged especially among family or household members. This solution going against it might be counter-intuitive to that.

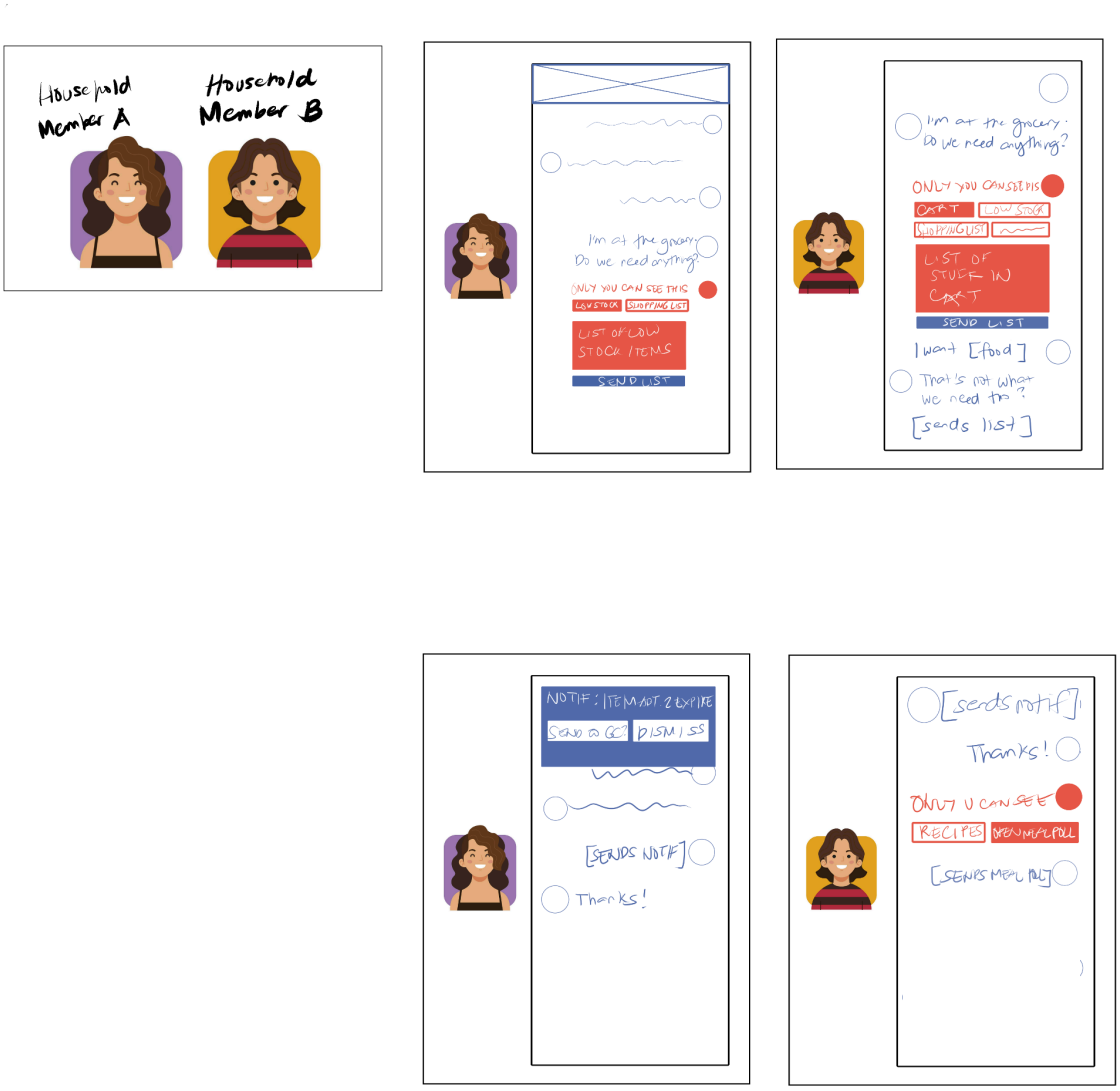
Key Insights

With further evaluation by myself, I realized that there might be more holes to poke with my idea. There is the problem that people who somehow do not have access to their smartphones or the NFC feature on their devices might just not be able to open their food storage. I needed to think of a way to allow users to still access it without their devices, but still hold people accountable. Furthermore, I have not fully researched the capabilities and limitations of NFC technology. There might be better alternatives or technical problems in it that I have not discovered yet. I think that targeting accountability was the right track for me to pursue, but I needed more research to fully polish my idea.

Storyboard 2 - Problem 2

Problem: Household has miscommunication or no communication at all regarding matters such as what to buy, what to eat, what has been bought, what is about to expire, stocks about to run out

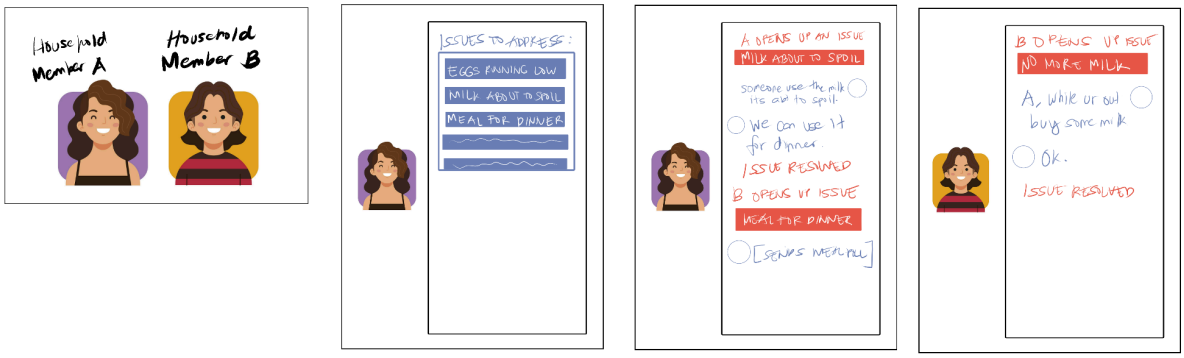
Solution 1:



In order to motivate users to collaborate on groceries, the solution I thought of would make it easier to have references on the topic at hand. The chatroom app would have a text detection feature. When users open or talk about a topic, it detects keywords and provides possible actions depending on the detected feature. For example, the keyword “buy” appears in the conversation; actions such as showing the shopping list, the low-stock items and the individual want items would appear. Initially, it is only seen by each user and not publicly shared to the

group chat but there will be an option to send it if the user deems it fit. The chatroom app also has a feature of sending meal polls to gather insights on what they would want to eat for future meals.

Solution 2:



This is another approach to motivate users to collaborate on groceries. However, this differs from the first solution by having a list of issues that the household needs to address. The users are now able to track what topics they need to talk about and they are also able to bring it up to the group chat. They can send or “open” an issue to the group chat and users are able to talk about them and resolve them. Additionally, it also has the same feature of sending meal polls in order to reach a consensus about future meals.

Evaluation:
During the group’s evaluation, I realized that these two solutions could be combined. However, there would be an issue in having too much going on. It could also have an issue of information being too hidden. Additionally, it was also pointed out that the meal poll felt out of place. This was because a lot of features showed task discussions and task tracking but this one did not live up to the other features as it only had a poll feature to reach consensus on the food to eat. Additionally, it was also suggested that the meal suggestions could be based on the ingredients in the pantry.

Key Insights:
Because of the evaluation, I was able to see perspectives I wouldn’t have originally seen. This made me learn that I should consider how each feature should complement one another. When there’s no complementation, the feature would be “incomplete”. If this happens, it would affect the efficiency of motivating users to collaborate with one another. Going forward, I would prioritize having information be well presented and organized. This is because having useful features is certainly helpful but if they are not properly presented, users would still have a hard time using them.

Storyboard 3 - Problem 3

Problem: Household has conflicts in preferences leading to wasted unpreferred bought items
Storyboard



The screen on the left-most side displays how one would be able to access the design which in this case is a shopping cart. The following two screens are examples of what groceries each person desires to buy. The one after that is an administrator view or the view for whoever is buying the groceries. They would be able to see a concatenated view of what each person in the household wants to get for groceries and they would be able to remove items that may not be in the budget or may not be good for the household giving them “veto powers”. The last screen on the right is what would appear after clicking on the hamburger button to access the different carts.

Narrative

The reason for choosing this option comes from the main point of the problem: conflicts. In my experience, the best or usually most straightforward way to resolve conflicts is to include an unbiased third party to allow the ones having conflicts to reach a consensus. In this case, the third party is the administrator or the one buying the groceries. The different people in the household will be able to add whatever they want without limit to the grocery list, but the one who is buying the groceries will have the final say on what will be bought. This will ensure that needs are bought and wants are not wasted.

Evaluation

After presenting the proposed solution, the first problem that arose was what would happen if people in the household added the same items to their shopping cart. The second was about the prices that are displayed on the shopping carts. It was pointed out that different groceries would have different prices for items and pricing is not always consistent. Aside from these, however, they believe the implementation to be a potential solution.

Key Insights

Based on the feedback provided, I am able to address the different issues to flesh out the solution even further. In terms of duplicate items in different shopping carts, it would be possible to provide multiple views instead that may either merge all the duplicates or separate them by color based on who added the grocery into the shopping cart. In terms of the price, it

may be more practical to remove it altogether and simply let the grocery shopper decide whether or not they want to keep the food items in the cart based on the prices they see in the grocery that they are in.

Storyboard 4 - Problem 4

Problem: Household does not know what food items are stored in their storage in real-time

Storyboard

The person forgets the food items stored in the fridge.

What do I have in the fridge again?



The person can check the app to know the food items in the fridge

Let me check the app



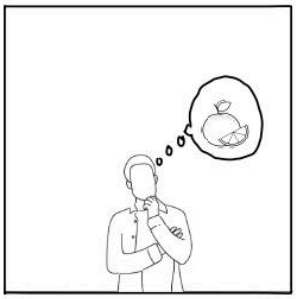
Upon opening the app, the homepage view will be shown.

Opens app showing homepage



If there are new food items that the person has purchased, the person will need to add that item to the inventory

There's still chicken, but didn't I bought 2 oranges, let me add that



The person can then access the storage after clicking the View Storage button

Clicks View Storage 1



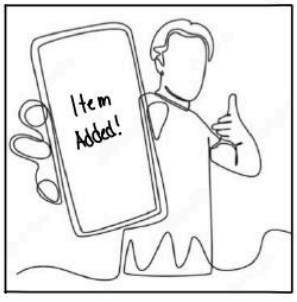
The person can access the add item page to add food items to their chosen storage.

Proceed to Add Item Page



The item is now added to the storage and will be shown when viewing the storage.

It's now added to the storage



Narrative

In response to the problem of households not being able to keep track of real-time food items in their storage, I thought of a practical solution. The idea was to develop a user-friendly app wherein users can input detailed information about food items stored in multiple locations, including quantity, name, expiration date, and other necessary information. This app would serve as a digital inventory for their food items, making it easier for them to manage and access this information when needed. This is the solution I thought of with the aim of simplifying the process of keeping track of food supplies.

Evaluation

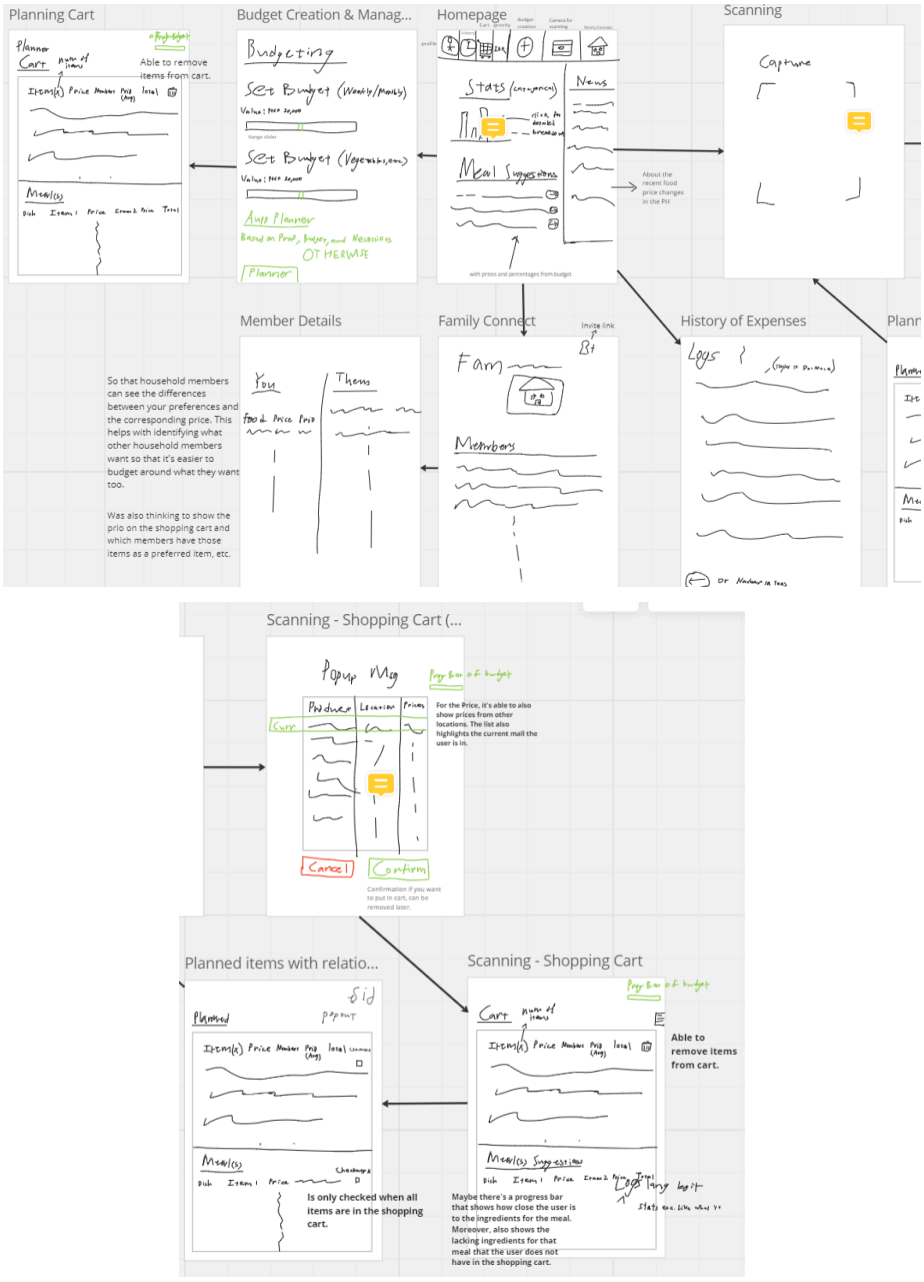
Upon presenting the proposed solution to my groupmates, I received feedback about the potential drawbacks of the solution. Some group members expressed their concerns about the user experience. They pointed out that the users need to first organize the food items on their storage, physically, and then invest the time to add the food items in the app. This is a two-step process which might be time-consuming and may discourage users from using the app. There was a consensus that while the app could be useful, the implementation potentially has practical challenges.

Key Insights

The feedback from my groupmates highlighted that the user experience and the practicality of implementing the solution must be considered. It became clear that the success of the app would depend on user behavior, and any pain points in the input process could deter users from consistently using the app. Based on this, my key insight is that the app's design and functionality should be streamlined to minimize the effort required from users. Simplifying the process of adding and managing food items, possibly through features such as barcode scanning of each food item, or scanning the receipt after grocery shopping, could significantly enhance its efficiency and overall user experience.

Storyboard 5 - Problem 5

Problem: Household has difficulty budgeting and prioritizing
Storyboard:



Narrative

As the difficulty of budgeting and prioritizing persists among households, especially in relation to the wants and needs of the households, a solution, in the form of an app, was formulated. The idea was to aid households in managing and planning their budgets and

priorities. Thus, my solution was an app that aided with such tasks, one that helps visualize and automate budgeting and prioritization.

In the images (solution) above, the app starts at the homepage and has multiple features, but it mainly shows important statistics categorically (e.g. which items were bought more), meal suggestions with price and percentages from the budget, and recent price changes to food items available in the Philippines. The app's navbar is where the main features for the app are located, namely Profile, History, Cart, priority, Budget Creation, camera icon for Scanning, and Family Connect.

The standard flow when accessing the app is to first utilize the Family Connect feature, where they can send invite links to people to become a part of their "Family", like a guild in games. The purpose of this is so that when you're connected to a family, you can access other members' details to see what the differences in food preferences are in order to get an idea of what to plan around.

Afterward, they can start budget creation (this also has an auto planner that optimizes the planning around priority and budget; no algorithm for this yet) and see the items in the planning cart. There are also meal suggestions to give ideas of what users can buy with the respective total prices required for the ingredients.

When users have planned what to buy (based on budget), they can go grocery shopping and scan the barcodes to get the item details in which a pop-up message should show up, confirming if they would want to add that to their cart and is also compared to other items from different branches/location/stores, etc. They are able to see the shopping cart and compare it with the planned cart to see if they are following as planned and to also see if they are over budget.

There is also a log for the expenses so that they can see whenever they went over budget or which category of items cost the most.

Evaluation

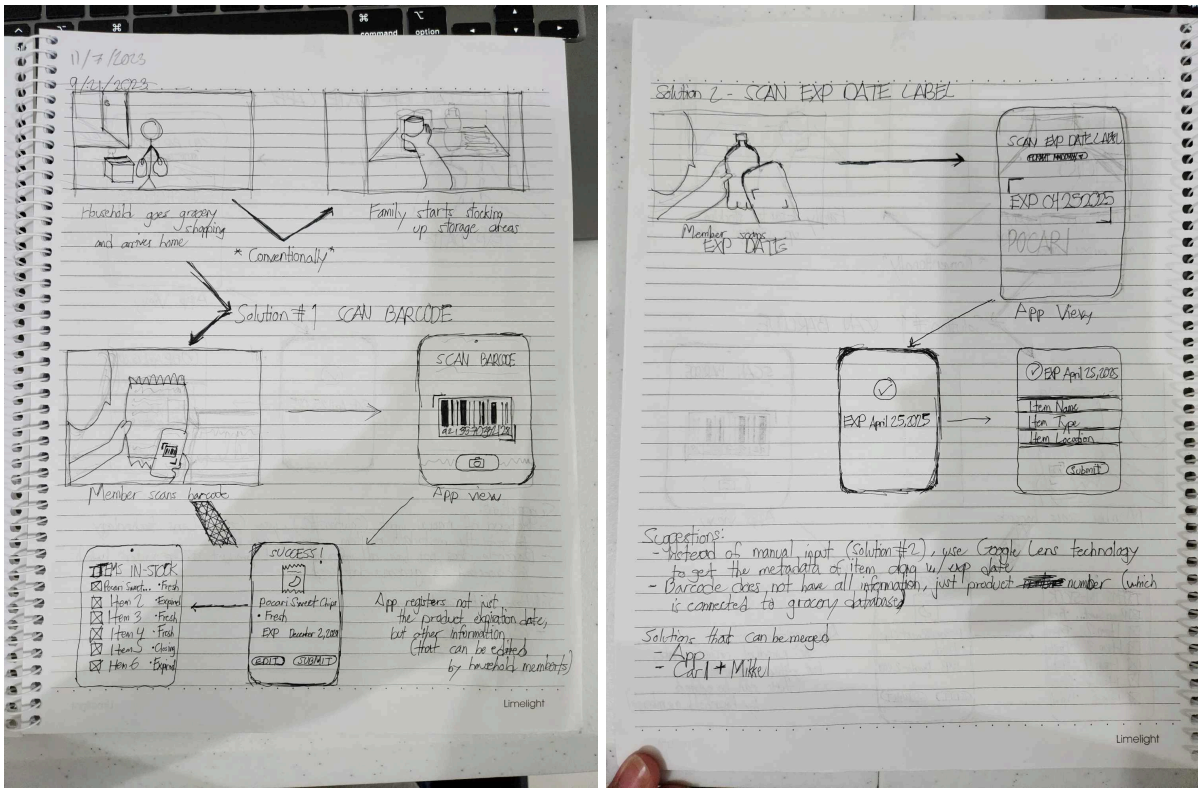
When the solution was presented, the main criticism was the lack of feasibility of the scanning of barcodes since it was brought up that the barcodes were more of an implementation that relied on the store's database. An alternative solution was proposed instead of barcode scanning, which was to implement the scanning of receipts after the user finished shopping. Another issue brought up was with regard to location since some people are not willing to go to other stores even if they have cheaper prices. Moreover, while there may be some locations where the user stays, the store they're going to is the only one around. While the app had a lot of good ideas, there were some minor issues, specifically the convoluted group of ideas just for a single problem.

Key Insights

It became clear to me after my group member's criticism that feasibility and a heavy focus on the problem are very crucial. Thus, by incorporating the suggestions presented by my group members of focusing more on a reactive approach in scanning receipts instead of a proactive one in scanning barcodes during the shopping phase and handling the issue with locations, the issues of the solution may potentially be addressed and lead to a more effective solution in the succeeding iterations.

Storyboard 6 - Problem 6

Problem: Household does not know the expiration dates of their food items



Description

The left picture provides a situation where a family arrives home after grocery shopping. Conventionally, they would go straight to storing and organizing the groceries into their appropriate storage areas. However, if a household wants to make sure that they can also keep track of their groceries' expiration dates, there are two proposed solutions.

The first solution proposes the use of an app that scans the barcode of a food item. This solution relies on the idea of the barcodes of food items containing all the essential information of the food item itself. Once successfully scanned, the app will show the following details of the scanned item: picture, item name, freshness status, and expiration date. The user then can edit the received data or submit it to the app's list of other scanned items. The full list contains every other scanned food item, their pictures, their names, and their freshness status.

The second solution, as shown in the right picture, is very similar to the first. Still, instead of scanning a barcode, it scans a food item's expiration date label, provided the user chooses the correct format (e.g., MMDDYYYY, DDMMYY, etc.). This solution relies on the idea of how most food items have their expiration date labels in similar formats and designs. Once scanned, the app will show a checkmark (indicating the label was successfully scanned) and the received expiration date. The app then prompts the user to add important information about the food item, such as its name, type, and location (where it is located/stored in the household's residence). This solution also utilizes the first solution's full list of other scanned items.

Narrative

The two solutions were developed with the idea that there needs to be a way to move the mental challenge of having to memorize and routinely check the expiration dates of food items away from individuals. There also needs to be a way to ensure that household members have accessible ways of checking the expiration dates of their food items. Finally, there also needs to be a way to keep track of expiration dates without manually inputting every expiration date of every food item at home.

Hence, the idea of an app-based solution seemed to be the way to go, given how almost all households have at least one member with a smartphone.

Evaluation

When the two solutions were presented to the rest of the group, the main criticism that was brought up was how there was a clear misunderstanding of how barcodes work with regard to the first solution.

To recall, the first solution relied on the idea that the barcode of a food item held all the relevant information of that particular food item, but apparently, barcodes simply contain the unique identification number for the product, which is also referred to as the UPC (Universal Product Code) of a food item. When a barcode is scanned, only the UPC is read and then sent to a store's database. The database then becomes the primary source of that food item's information, such as its name and expiry date. This newfound understanding renders the first solution completely useless.

The main suggestion that was given and deemed to be applicable for both solutions was the use of Google Lens technology in order to input the food item's relevant information automatically. For example, if it were to be implemented along with the second solution, instead of the app's scanner scanning a particular food item's expiration date alone, it could also take pictures of that item's packaging. The photos can then be processed to automatically get all the necessary information about the given food item. No further comments or suggestions were brought up.

Key Insights

Despite the criticism and need for improvement especially on the research side, the evaluation of the two solutions provided by the group indicated that the proposed solutions are moving in the right direction. By properly addressing the misunderstanding of barcodes and incorporating the suggestion of adding technology similar to that of Google Lens, the next iteration of solutions has the potential to effectively address the struggle with households and their food items' expiration dates.

Checkpoint 2.1 (Merging of the 6 solutions into 3)

Brainstorming

The group started the meeting to discuss what to do for checkpoint 2 with the main goal of merging the 6 initial solutions into 3 solutions to slowly get closer to having 1 final solution.

It was decided that since Problem 1 had proposed multiple solutions, they would stay unmerged with others' solutions and instead merge all their initial solutions into one. For the rest of the group, those assigned to problems 3, 4, and 5 agreed to merge their solutions together as they had very similar solutions that could work hand-in-hand to solve their problems. Finally, those assigned to problems 2 and 6 merged their solutions in an effort to help each other improve upon each other's solutions and understand possible ways of merging dissimilar solutions.

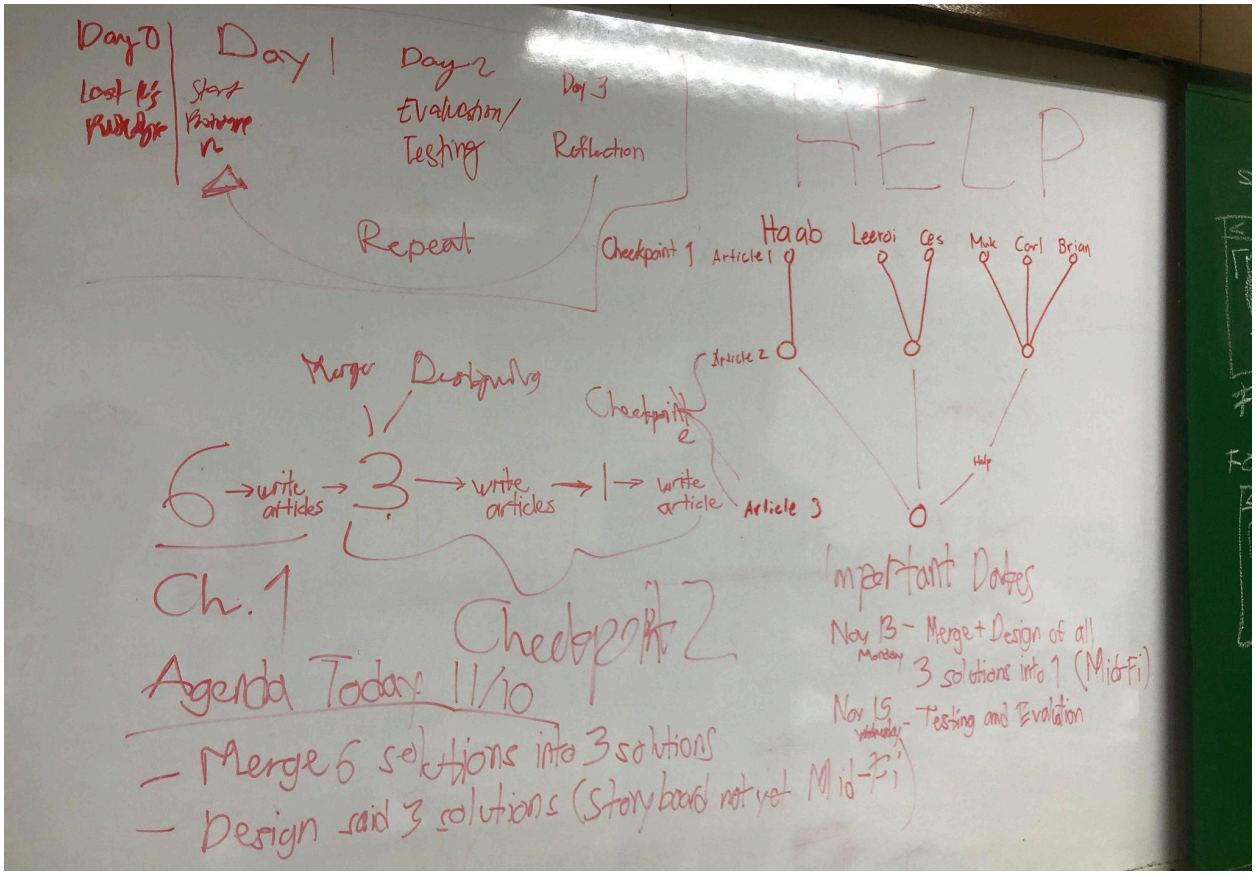


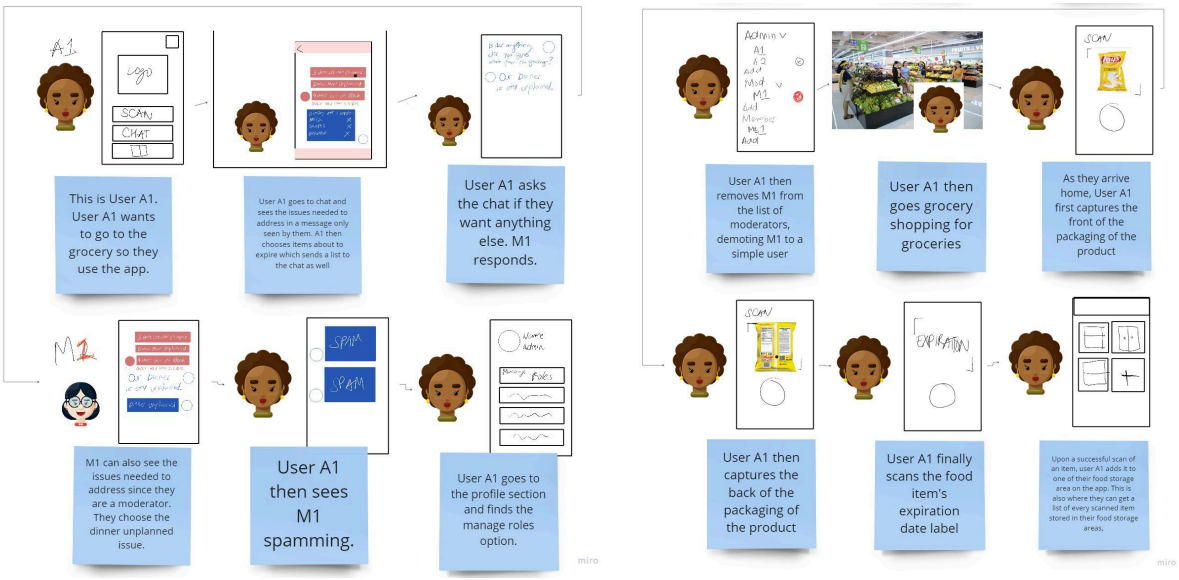
Figure 4. The board during the meeting

After this discussion, the members of the group went to work conceptualizing and designing the 3 solutions.

Merged Solution 1 - Problems 2 and 6

Problems

- Household has miscommunication or no communication at all regarding matters such as what to buy, what to eat, what has been bought, what is about to expire, stocks about to run out
- Household does not know the expiration dates of their food items



Description

The picture on the left side shows part 1 of the scenario, while the right picture shows part 2 of the scenario. The frames are read from left to right per row. Suppose User “A1” (which stands for Admin 1) wants to go to the grocery. So they check the app to see what needs to be bought. They go straight to the chatroom feature and a list of issues that need to be addressed in the household (only seen by the current user and not shared publicly with the chat). They click the issue that’s focused on the items that are about to expire. When they click that issue, it is now sent to the chat as a list. Another member of the household, M1 (stands for Moderator 1), sends a message to the chat saying that their dinner is still unplanned. On their side of the screen, they also see the list of issues that need to be addressed while still not being seen publicly in the chat. They then click the Dinner Meal unplanned issue. However, after that, they begin spamming the chat by clicking several issues that have no connection to the topic. A1 then manages this by going to the profile section and finding the manage roles option. With that, they demote M1 from a Moderator to a Member. Each role has restrictions that are managed by the Admin.

A1 then goes to the grocery. After doing so, A1 proceeds to take a picture of the front and back of the food item’s packaging. They also scan the expiration date of the food item. Subsequently, A1 is now able to add the food item to one of the food storage areas of their app. This part of the app allows users to see the list of food items stored in their food storage areas.

Narrative

The solution was made by combining the initial solutions of two members from the previous iteration. As both of these members had two solutions initially, one solution was picked from each after a thorough discussion. For the collaboration part, the solution of having an issue tracker was picked because if users rely too much on the text detection feature, they would not be able to remember the issues they need to talk about. For the expiration date part, the solution of scanning the expiration date was chosen because scanning the barcode is not feasible as the barcodes are implemented based on the grocery the food item was bought from. Additionally, the scanning of the expiration date solution was further improved upon by taking a picture of the whole item itself using Google Lens technology. With that, the user would no longer need to input the details of the food item one by one.

With the combined solution, we were able to create a solution that would allow household members to collaborate on tracking and remembering the expiration dates of the food items in the household.

Evaluation

When the two solutions were presented to the rest of the group, the first comments made were about how other sub-groups in the team had very similar solutions for the storing of food items on the app, where all sub-groups' implementations of the virtual food storage areas have the potential to be merged into one.

However, there were doubts about the scanning functionality, specifically the scanning of the expiration label, since there exist food items whose expiration date labels are too illegible that the scanner might not pick the text up, especially food items whose expiry labels blend in too well with their packaging that you have to shine a light on the expiry label in a certain way just to be able to see the expiry date properly. However, the rest of the group did admit that it seems like there is no other solution that allows for a more seamless detection of expiry date labels. There was a suggestion that was brought up to allow for the manual checking of expiry dates of a household's food items. Each week, a different household member would be assigned to manually check on each food item's expiry date and inform the household about items that have either expired or are close to expiring. However, this suggestion was immediately shut down as it was thought that it would not work in the long run without a proper incentive for the household members to keep doing their weekly tasks.

One criticism made towards the solution was in regards to the scanning feature, where its current implementation only really seems to cater to food items with a proper front and back side to their packaging. For example, this would mean that it would be easy to add chips to the virtual food storage since their packaging has a definite front and back, but the same cannot be said for canned goods like corned beef.

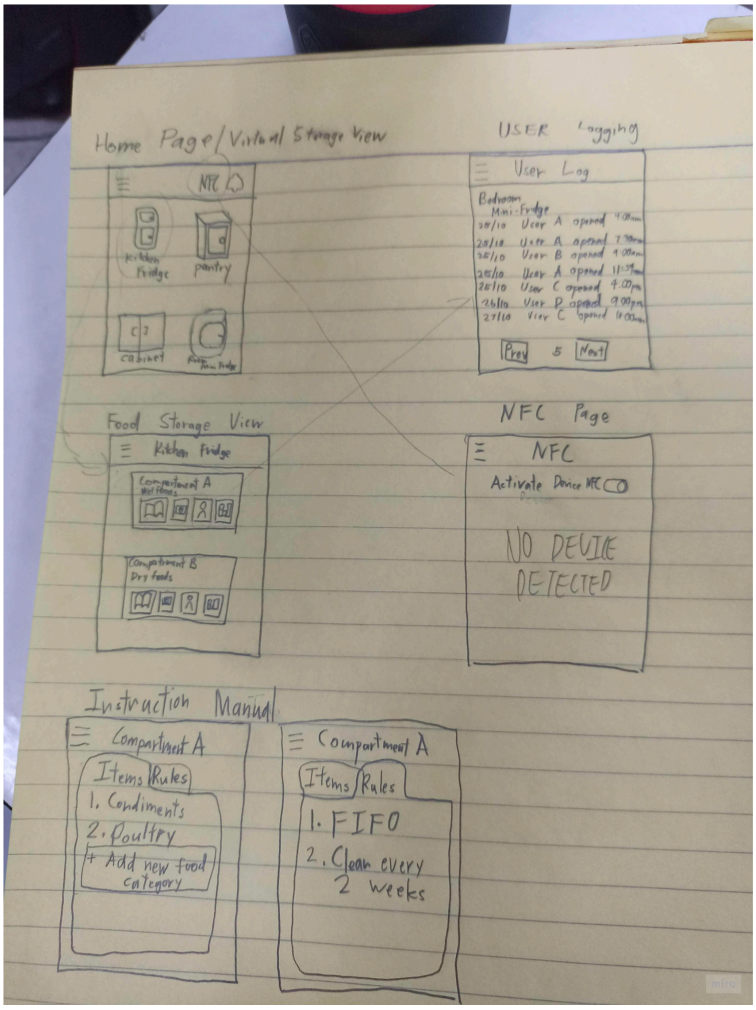
A minor comment for the chat moderation feature was made in that it should act as an extra feature in the grand scheme of the entire solution.

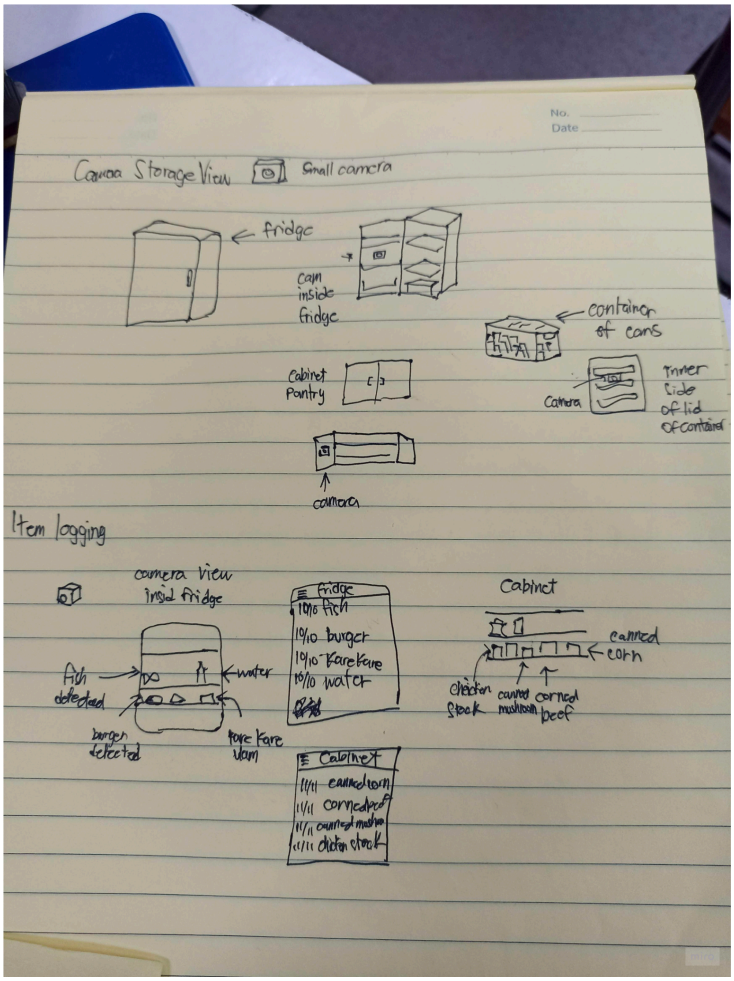
Key Insights

Overall, despite how tedious it may be, the rest of the group agrees that the scanning of the expiration date label is an essential part of the solution as it solves the problem of having to manually type every food item's expiry date label. They also agree that the collaboration-via-chat feature was well-implemented and well-designed, given its connection with the app's virtual food storage.

The evaluation given by the rest of the group highlights the solution's strong and weak points, with one notable strong point being its similarity in design and implementation with the solutions of the other sub-groups. Understanding these points individually will help us make more informed decisions about what features of the solution can be brought over and merged with the other solutions, along with what can be omitted to create a useful and user-friendly final solution.

Merged Solution 2 - Problem 1





Narrative

Kept Ideas

We unanimously agreed to retain NFC Identification and Virtual Food Storage for the convenience of utilizing an app to access and manage the contents of household storage instantly. This eliminates the need to search through each storage area manually. The NFC identification feature also allows us to maintain detailed logs for each household member, tracking their contributions, organization efforts, and consumption from specific storage units, which helps in terms of the collaborative work of each household.

New Features

We added and changed existing features that we think would add to seamlessness and convenience for our users. With regards to the issue of adding items to the virtual storage and requiring double work, we changed it into a general instruction manual or guidelines customized by the users themself. Users may define in each storage compartment the general items that should be in them, how the items must be arranged, and even details on when and how they should be cleaned. The next feature we added was a camera storage view, and this allows items logging and real-time viewing of the storage. The item would be identified through image recognition, and if it doesn't fit the general items that were assigned by the user in the instruction manual, it would alert and notify the user that an incorrect item was placed. The next key feature is interconnected with the camera feature as it allows item logging. In our initial feature, we only had user logging, but with the storage camera feature, we would also be able to log the items placed in the compartment.

Evaluation

Camera Issues

The notion of a camera possessing the ability to instantly identify food items at a glance and subsequently log them into respective food storage areas is disregarded due to the current limitations of AI. The technology has not reached a level where it can perform such functions. Furthermore, the development of a clear, efficient, and sustainable camera for this purpose proves to be costly. Even with a high-end camera quality, this solution fails to address the diverse blind spots inherent in various types of food storage containers.

NFC Locking

An issue encountered with the NFC locking was realized last checkpoint which was, "How will people access their food if they don't have their current device that allows NFC technology". The team discussed ideas that would work in either changing the idea or having a band-aid solution for this problem. I proposed that smart fridges that have touch screens can allow users who forgot their NFC device to type their name before allowing the fridge to be opened, but these users who opened the fridge this way would be tagged as guest users. Other ideas they asked me to explore were fingerprint recognition and facial recognition.

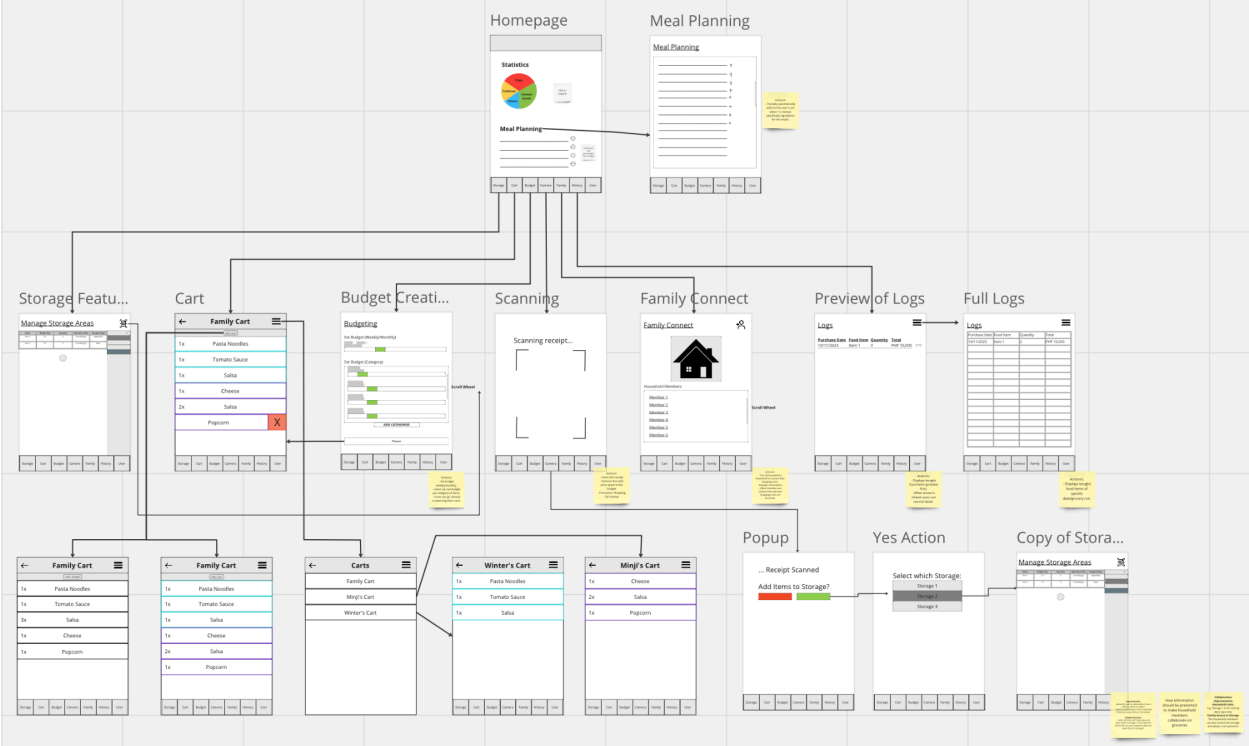
Key Insights

The verdict of our members was that we had good core ideas, but there were issues in practicality within them. We realize that utilizing AI as a cover term to use solutions we didn't know how to solve was to be avoided (This is specifically for the camera item identification feature). Accountability is still considered vital to solving the problem of double organization; furthermore, its byproduct effect of mindfulness is what we should aim for. In merging our ideas, it was easy to be idealistic and forget the logistics of the solutions we were formulating, which we think is something to consider when collaborating with others.

Merged Solution 3 - Problems 3, 4, and 5

Problems:

- Household has conflicts in preferences leading to wasted unpreferred bought items
- Household does not know what food items are stored in their storage in real-time
- Household has difficulty budgeting and prioritizing



Description

The first frame that was made consists of the homepage, which sits above the rest. This frame includes statistics on financial expenses relating to grocery shopping, a meal planning option that leads to a different frame allowing the household to add specific meals into their plans, and lastly, a quick action bar that's found in almost every frame. The bar consists of the following: "Storage", "Cart", "Budget", "Camera", "Family", "History", and "User". Following this order, the storage option includes the ability to go through the different storages of the household in the sidebar on the right and the ability to see its contents on the table in the middle. Next, the cart consists of the planned grocery items of each household member and can be viewed either based on the desires per member or based on an aggregated collection of all their wants. Thirdly, the budget section allows the user to set their preferred budget by how often they grocery shop or on different grocery categories such as "Vegetables", "Protein", and "Toiletries". After that, the camera option is present to provide the option of scanning receipts allowing users to automatically include their bought groceries into the storage system. The family section comes next which is where the option to include family members in the system is located as well as providing administrative powers to the one who will have a final say on the grocery list. Lastly, with the user section not having a frame yet, is the history section which logs

every grocery that has been bought including the total and individual prices, the date they were purchased, and its corresponding quantity.

Narrative

It was decided to keep some of the features, namely the family connect feature that helps with collaboration amongst household members with regard to their budget, the idea of having different shopping carts per household member, and the idea of having multiple virtual storage areas inside the app.

However, from the previous checkpoint, there were some improvements that were made. We essentially just took into consideration all the concerns last time and improved upon them. Just as we took into consideration if there are duplicate items, now they're combined among users, and the number of total items desired by the household members are combined. Prices displayed previously for the household members' carts were removed due to the previous concerns about different stores having different prices and are not always consistent. The user experience issues were mitigated as well, specifically the two-step process for storing food items that initially required physically storing the items and also manually inputting them digitally. This was achieved by making the app able to scan receipts and then add all the items there to the virtual storage, which also addresses the issues of the infeasibility of scanning barcodes with scanning receipts instead for food items. Instead of just having one Prices of food items based on store branches were also removed since not everyone has alternatives based on location or does not prefer them. Moreover, a lot of other variables (such as the availability of stores in their location, varying prices per store, etc.) would also have to be taken into consideration if locations were to be implemented and thus was deemed infeasible and removed. The aforementioned ideas were essentially just merged, improved, and fleshed out to make a more efficient version of the prior solutions.

Evaluation

During the group evaluation of the second prototype, my groupmates provided valuable feedback on the new features, particularly the scanning and storage area management functionalities. They positively acknowledged the scanning feature's potential to simplify user input by scanning receipts and efficiently adding food items to storage areas. However, there was a noted limitation in the current prototype, where scanning a receipt stores all items in a single location. To enhance management capabilities, a proposed modification includes adding a select and filter feature to choose multiple items for placement in a selected storage area. Additionally, if an existing item is present, users should be prompted to decide whether to overwrite the stored item or add to its quantity. The group's feedback underscores the significance of user experience, and the proposed adjustments aim to address these concerns appropriately.

Key Insights

The features that were once vague kept improving because of rapid prototyping and brainstorming from the group. The insights from the evaluation emphasized the importance of user flexibility and control in the app's functionalities. Providing options for manual selection

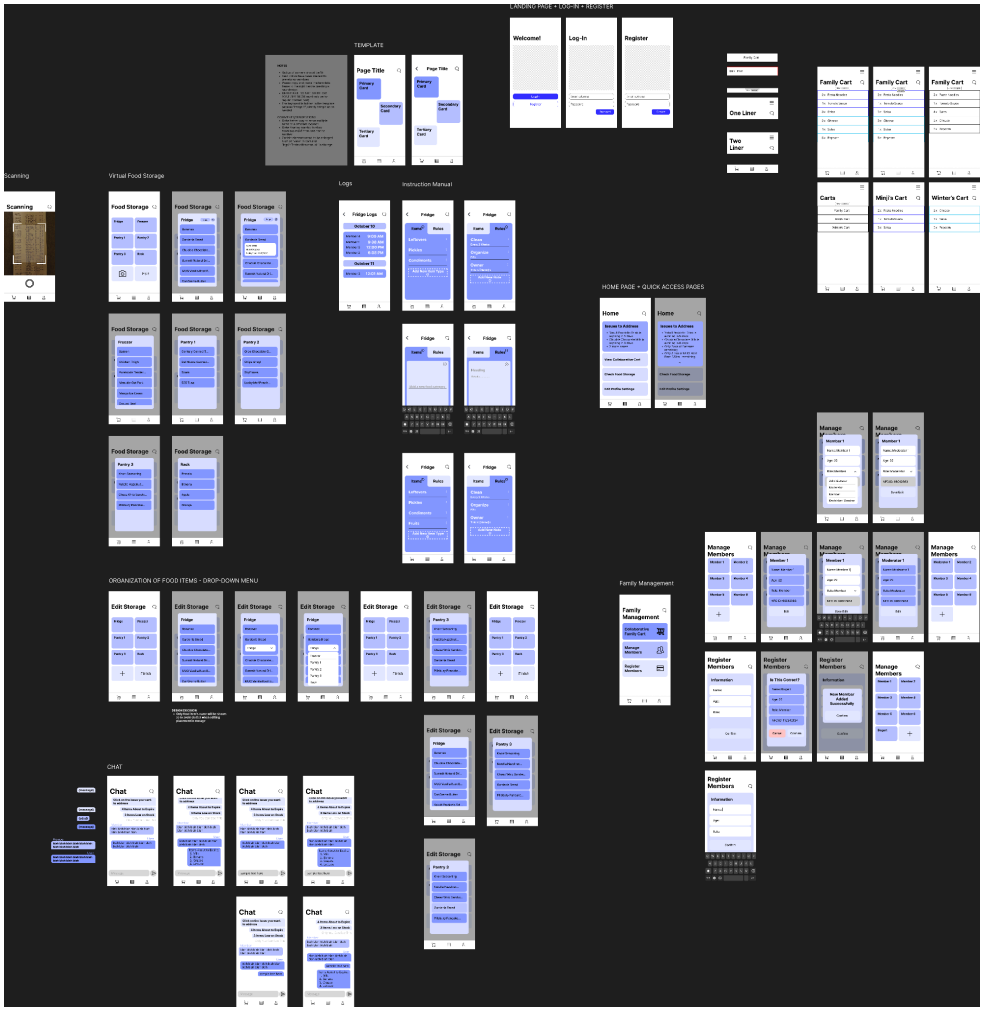
after scanning and allowing users to choose storage locations reflected a user-centric approach to the design. Additionally, addressing conflicts during the input process when overwriting existing entries or adding quantities to existing stock is a key consideration. The integration of these insights into the application aligns with the goal of creating an efficient and streamlined experience for users.

Checkpoint 2.2 (1st iteration of mid-fidelity prototype)

Brainstorming

After the evaluation of the 3 merged solutions, the group started discussing the first few steps of coming up with the final solution prototype. We first had to figure out which of the three solutions would be used as the “foundation” where the other two solutions would simply be added. Ultimately, it was decided that Merged Solution 3 would serve as the foundation since their solution seemed to be the most fleshed-out.

We then decided whether we create a mid-fidelity prototype or go straight to creating a high-fidelity prototype. By split decision, we decided to go with a high-fidelity prototype using Figma since it seemed like a waste of time and effort to create a mid-fidelity prototype using Miro, which in terms of prototype creation, essentially functioned the same way as Figma. Each member was assigned to create the Figma designs for their own respective problem from the first iteration of solutions (the six solutions).



Usability Testing

Once the first iteration of the final prototype was completed, we decided to evaluate the design through usability testing. The group came up with the following prompts for our potential testers:

1. Log In to get to the home page
2. Find the grocery items in “Minji”’s cart
3. Give Member 1 the “Moderator” role
4. Add a new member named “Bogart”
5. Try adding new items to your storage by scanning a receipt
6. Check the details of the Gardenia Bread in your food storage
7. Move the Gardenia Bread to the “Pantry 3” storage area
8. Try bringing up an issue in the chat
9. Try sending a message in the chat

The results of the usability test were recorded and stored on a Google Sheet, which included the errors the users made, their errors per second, their completion time, and more. As of November 16, 2023, there have been 2 volunteers for the usability testing.

User 1:

	Total Users:	2					
USER 1							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
log in to get to the home page		0	8				TRUE
Try and find the items in “Minji”’s cart	iiiiiiiii	9	82			Hamburger icon not fit for feature, view dropdown misleadingly way to switch carts	FALSE
Give Member 1 the role of a moderator		0	13				TRUE

	Total Users:	2					
USER 1							
Add a new member named “Bogart”	i	1	13				TRUE
Try adding new items to your storage by scanning a receipt		0	4				TRUE
Check the details of the Gardenia Bread in your food storage		0	13				TRUE
Move the Gardenia Bread to the “Pantry 3” storage area	iii	3	42			Edit button/function better off being part of the main food storage view	TRUE
Try bringing up an issue in the chat	iii	3	17				TRUE
Try sending a message in the chat		0	5				TRUE

User 2:

USER 2							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
log in to get to the home page	ii	2	38			add mock email and address; need to register daw	TRUE

Try and find the items in “Minji”’s cart	iiiii	5	48				TRUE
Give Member 1 the role of a moderator	ii	2	24				TRUE
Add a new member named “Bogart”		0	15				TRUE
Try adding new items to your storage by scanning a receipt		0	17				TRUE
Check the details of the Gardenia Bread in your food storage		0	11				TRUE
Move the Gardenia Bread to the “Pantry 3” storage area		0	12				TRUE
Try bringing up an issue in the chat		0	9				TRUE
Try sending a message in the chat		0	5				TRUE

The results of our two usability tests have been generally promising, with a reasonably high completion rate at a short amount of time for most of the tasks that were tested. But despite these favorable results, we learned that a few things need to be changed for the next iteration of the design.

The first is specifically for the Family Cart feature, where you could check the items in the carts of other members. We found that the testers were generally confused with where they could find this information as the navigation symbols and buttons used in the app were not fully representative of their actual function.

The second change that must be made is with the editing of the food items in the virtual storage areas. It was pointed out that the current design does not allow the moving of multiple food items to one food storage area.

Finally, the third change that must be made is to the color palette. The app currently utilizes a single-color palette, with the different elements being different shades of blue.

Unfortunately, one of the shades frequently used in the app is a bit too light and seems to easily blend in with the actual background of the app.

Checkpoint 3.1 (2nd iteration of mid-fi prototype)

Brainstorming

The group started the brainstorming session by first discussing the criteria of the meeting, specifically what needs finalizing and polishing with regard to the app-based solution. These criteria were based on feedback from our professor upon presenting the most updated solution we had at the time.

STHCIUX - MCO2 Checkpoint 3/Final Checkpoint

GDRIVE LINK: https://drive.google.com/drive/folders/1xy_8bL-V139AUOpiKqRcjnr-dm_jkX_s

TOPIC: Finalizing/polishing of final solution

- Move back to lo-fi unless you can fix the mid-fidelity prototype quickly
- Lessen the scope of tasks to only specific functions that you want to test
    - Focus on tasks with tons of error during first usability test
- Color palette should be the most minor problem, only fix at high-fidelity stage
- Focus on finding testers from different backgrounds such as different ages, different economic classes, different household setups (nuclear, extended, solo, etc.)
    - Different ages
        - Parents/adults in the household
        - Younger siblings
    - Different economic classes
        - Get past interviewees as testers of app
        - Explicitly mention target demographic as explanation for only testing on a few economic classes
    - Different household setups (nuclear, extended, solo, etc.)
        - Get past interviewees as well
        - Group separation
            - Nuclear - Everyone else
            - Extended - Ces and Gab
- Take into account the SIZE of food storage areas
    - Measuring using phone's camera of storage space and food items
    - Let user put limit on the number of items that can be stored
    - Let app give warning to user when it recognizes a few too many items categorized/put into virtual food storage area
    - MERGED SOLUTION:
        - Implement limit (can be changed) on number of items that can be stored.
        - Users can go past that limit, which will make the app show a warning to the user
- Solution to already-implemented feature: Add virtual storage area for uncategorized/ambiguous food items after scanning receipt
- Add more functionality/screens to the scanning feature in Figma

Figure 5. The Google Docs document used to keep track of the meeting agenda and notes

Our group first discussed whether or not we should move back to lo-fi from mid-fi prototypes. This was brought up as a suggestion as to better facilitate larger changes in functionality with the app. Upon further discussion, the group unanimously decided to just stay on mid-fi since the group determined that we were able to quickly make adjustments to the mid-fi prototypes. The group then discussed ways of lessening the scope of the tasks that usability testers would be asked to perform. We decided that for the next few usability tests, we will be focusing more on the tasks that had the most errors during our first usability test. While there was an issue with regard to the color palette used for the prototype, the group has decided to defer settling that issue until we reach the high-fidelity stage and instead focus more on the features of the app and the general user experience. Another feedback we received was to take into consideration the different demographics of the testers, namely those of different backgrounds, age groups, economic classes, and household setups. This is in order for the app to be accessible to every type of household that could require the services of our app.

The group then discussed how we could get usability testers for the next set of testing of the updated prototype. For testing on the different age groups, we decided that it would be feasible to have our younger siblings or our parents in our households test the updated prototype. For the different economic classes, we decided to explicitly mention our target demographic as to why there are only a few economic classes included as testers. Lastly, we determined that it'd be hard to get testers from different households especially while knowing their different household setups, we decided to get past interviewees as well since they already mentioned their setups in the interviews. We first looked into the family groups that our group had, and segregated accordingly between nuclear and extended, while not considering solo since those aren't part of the group's target demographic. The significance of determining the household setups is so that we'll be able to determine whether or not the app's scalability can handle all types of households and not just small families.

The group also discussed the importance of taking into account the size of the food storage area. One member of the group suggested using the phone's camera to measure the storage space and food items, as well as allowing users to set a limit on the number of items that can be stored in a specific storage area. In addition, it was suggested that the app should give users a warning when it recognizes that there are too many items in a specific food storage area.

Finally, we have reached a consensus on a solution that allows users to adjust the limit on the number of items they can store, accommodating varying household storage practices. This flexibility enables users to pack items as tightly or neatly as they prefer, with the app serving as both a gauge and helper. The app will notify users when their food storage exceeds the assigned limit, promoting awareness. Additionally, we propose enhancing the scanning feature on the prototype by introducing a virtual storage area for uncategorized or ambiguous food items after scanning receipts. This improvement aims to make the functionality and screens of the scanning feature more intuitive for users.

Usability Testing

Once the second iteration of the final prototype was completed, we decided to evaluate the design through usability testing. The group came up with the following prompts for our potential testers:

1. Log-in to get to the home page
2. Try and find the food items in “Winter”’s Cart
3. Try checking the details of the Gardenia Bread in your food storage
4. Try moving the Gardenia Bread and MB Cheese Bites to the “Pantry 2” food storage area
5. Bring up the “4 Items About To Expire” issue in the chat

The results of the usability test were recorded and stored on a Google Sheet, which included the errors the users made, their errors per second, their completion time, and more. As of November 22, 2023, there have been 3 volunteers for the usability testing.

User 1:

USER 1							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
Log-in to get to the home page		0	8		0	recurring comments of why would u login agad without registering HAHA	TRUE
Try and find the food items in “Winter”’s Cart	iiiii	5	90	90	0.05555555556	add button sa main family storage for looking in indiv carts	FALSE
Try checking the details of the Gardenia Bread in your food storage			26		0		TRUE

USER 1							
Try moving the Gardenia Bread and MB Cheese Bites to the “Pantry 2” food storage area			12		0	do not assume that all people know how to select stuff	TRUE
Bring up the “4 Items About To Expire” issue in the chat			13		0		TRUE

User 2:

USER 2							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
Log-in to get to the home page		0	7				TRUE
Try and find the food items in “Winter”’s Cart	iiiiiiiiiiiiiii	18	90	90		"masyado maliit pindutan sa taas"	FALSE
Try checking the details of the Gardenia Bread in your food storage		0	42			"bat naman nasa ref yung tinapay"	TRUE
Try moving the Gardenia Bread and MB Cheese Bites to the “Pantry 2” food storage area		0	39				TRUE
Bring up the “4 Items About To Expire” issue in the chat		0	5				TRUE

User 3

USER 3							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
Log-in to get to the home page		0	6				TRUE
Try and find the food items in “Winter”’s Cart		0	21				TRUE
Try checking the details of the Gardenia Bread in your food storage		0	18				TRUE
Try moving the Gardenia Bread and MB Cheese Bites to the “Pantry 2” food storage area		0	15			hindi clear yung select checkmark	TRUE
Bring up the “4 Items About To Expire” issue in the chat		0	10				TRUE

Based on the usability testing conducted on our app with the help of 3 volunteers, the results indicate a high success rate and efficiency for some tasks, with 4 of the 5 tasks having a 100% completion rate. Logging in to access the home page demonstrated excellent usability, with a 100% success rate, zero errors, and a swift average completion time of 7 seconds. Similarly, checking details of an item in the food storage and moving items between storage areas both showed optimal performance, with 100% success rates, no errors, and efficient completion times. Another feature with favorable metrics was the task tor bring up the “4 Items About to Expire” issue in the chat, and it had a 100% success rate, no errors, and a quick completion time of 9.33s.

However, the task of finding food items in the “Winter” cart yielded a lower success rate of 33.3% with an average of 7.67 errors and a considerably longer average completion time of

111 seconds due to the challenges faced by 2 of the 3 volunteers. This task presents a notable usability challenge and may require further refinement and reimplementation/redesign from the group. The users voiced out their opinion that the buttons to navigate whichever cart to go to is too small to press. Another suggestion was to add an extra button to navigate individual carts from the family cart itself.

While certain aspects of the app exhibit excellent usability based on the tests done, the findings suggest that there is an issue regarding the task related to finding items in the “Winter” cart. There are also features that can be improved upon even if the results to other tasks are positive, such as redesigning the user interface for sending issues through chat, to allow better visibility of issues related to storage quantity being low for example.

Checkpoint 3.2 (3rd iteration of mid-fi prototype)

Brainstorming

During the brainstorming meeting, the group first discussed the different issues that the previous users encountered during the testing period. As mentioned in the previous article, the two main issues that were prevalent during the tests were navigating to a user's cart and editing the location of food items.

Considering that the same problem has been brought up multiple times already regarding navigation, inspiration was taken from different designs online to come up with an appropriate solution to the problem. As such, the new implementation for navigation functions similarly to "Stories" on Instagram or "Chat Heads" on Messenger. In terms of the confusion on the merged and unmerged items on the cart, it was discussed that the feature worked somewhat like a switch, so the new design was implemented using a toggle button.

Finally with the editing of the location, a comment from the users before mentioned that the circles for the checkmark on the storage were unintuitive. Hence they now only appear when the edit button is clicked. After discussing the problems and the potential solutions, the group figured out the new set of users that would be used for testing.

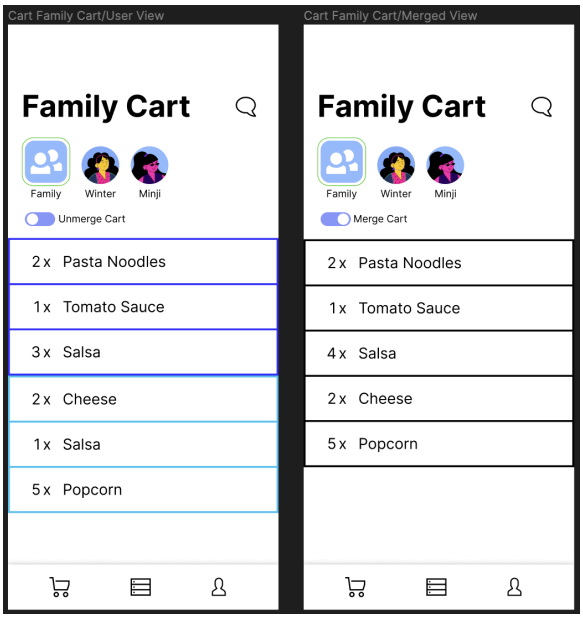


Figure 6. Cart Navigation; Merge/Unmerge

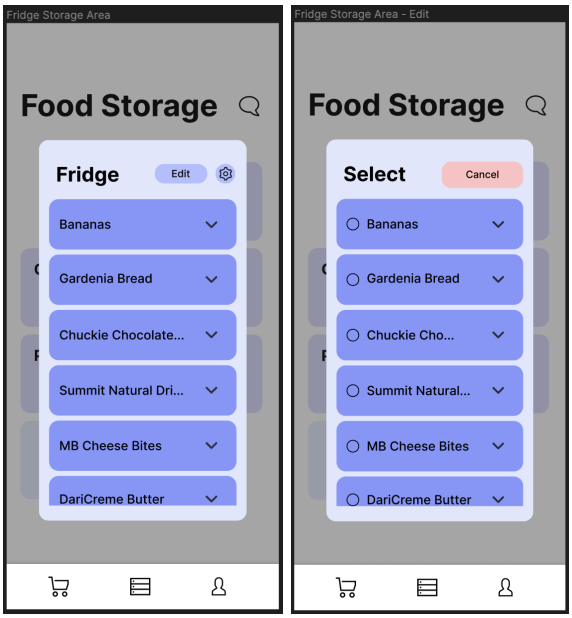


Figure 7. Edit food item location

Usability Testing

After the third iteration of the final prototype, the number of usability tests was limited to three as seen below:

1. Try and find the food items in “Winter’s” Cart.
2. Try moving the Gardenia Bread and MB Cheese Bites to the “Pantry 2” food storage area.
3. Register a new member to the household: Name is Bogart, 22 years old, with the role of a member.

The results were once again listed on Google Sheets and 3 different people from different households were taken for this test.

USER 1							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
Try and find the food items in “Winter”’s Cart.		0	8				TRUE
Try moving the Gardenia Bread and MB Cheese Bites to the “Pantry 2” food storage area.		0	21				TRUE
Register a new member to the household: Name is Bogart, 22 years old, with the role of a member.		0	18				TRUE

USER 2							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
Try and find the food items in “Winter”’s Cart.		0	30			Tutorial screens/More descriptions placed somewhere on the screen for better navigability. Flow is not immediately recognized for new users	TRUE
Try moving the Gardenia Bread and MB Cheese Bites to the “Pantry 2” food storage area.	i	1	70				TRUE
Register a new member to the household: Name is Bogart, 22 years old, with the role of a member.		0	47				TRUE

USER 3							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
Try and find the food items in “Winter”’s Cart.		0	20				TRUE
Try moving the Gardenia Bread and		0	20				TRUE

MB Cheese Bites to the “Pantry 2” food storage area.							
Register a new member to the household: Name is Bogart, 22 years old, with the role of a member.		0	30				TRUE

Based on the results seen above, the changes that were implemented were sufficient and appropriate for the different tasks, leaving the tests with a 100% completion rate on all tasks across the 3 users. All but one of the tasks were completed below 60 seconds which means that the changes were effective as well. However, based on the comments from the second user, improvements can still be made to refine the flow of the different screens and make it more intuitive for different users. All in all, aside from the potential tweaks to improve the application, it’s clear that different features can now be tested in a more in-depth fashion and we can now move on to testing different features of the app.

Checkpoint 3.3 (4th iteration of mid-fi prototype)

Brainstorming

The brainstorming session for this checkpoint started its discussion with the one error that occurred during the previous usability testing. After addressing the need to add more information for better navigability, the group decided to move on to a new set of usability tests.

Going through the Figma designs and identifying what else was missing, the group decided to test the logs, the rules, and the scanning features. Since there were still missing screens for the scanning feature during the first usability, its testing was prioritized for this iteration as additional screens were added that fleshed out the feature and how the group originally envisioned it to work. Complexities were also added to the test for scanning so that in one scenario, users would just have to add the items they scanned from the receipt directly into their storage without problems, while in the other scenario, users would hit a snag wherein the limit of the recommended storage is reached. This is why the group implemented an extra virtual storage called “Others” to temporarily store the items that no longer fit the recommended storage space.

With regards to the logs and rules, the design was updated to make it more user-friendly, while also adding new features, such as a new intermediate menu to allow the user to choose where to go among the logs, items, and rules screens. Other functions were also added, such as the adding and deleting of a storage area’s individual items and rules that were not implemented nor tested during the first usability test.

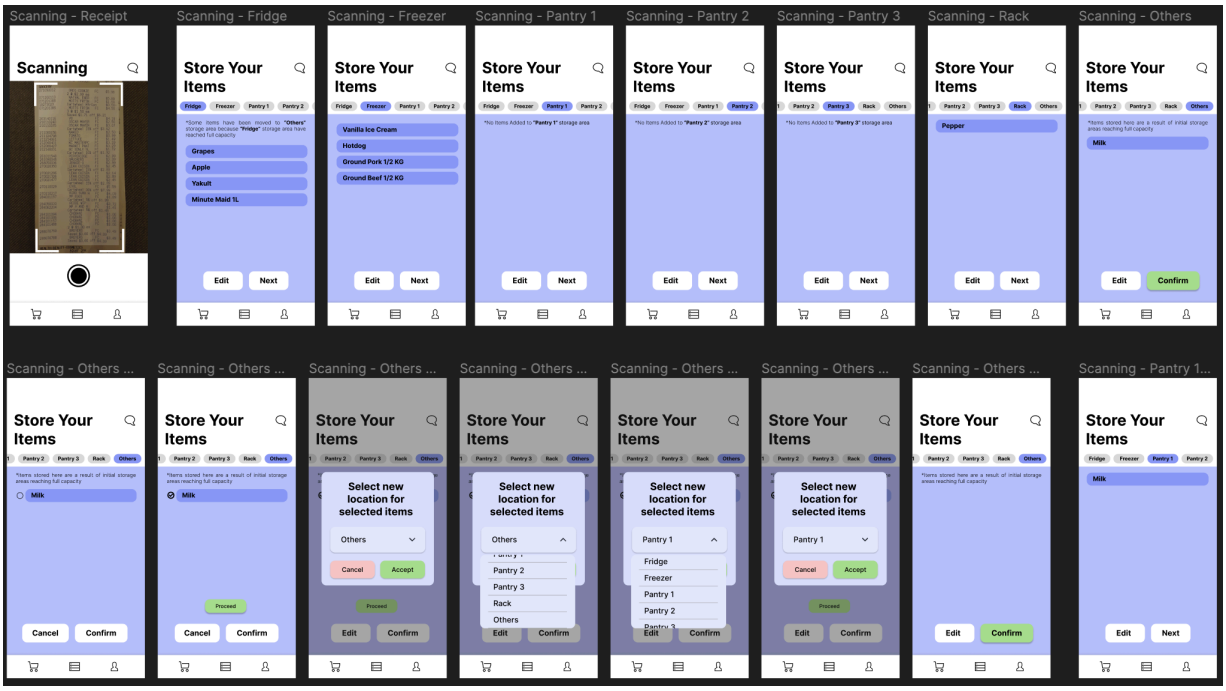


Figure 8. Scanning Screens

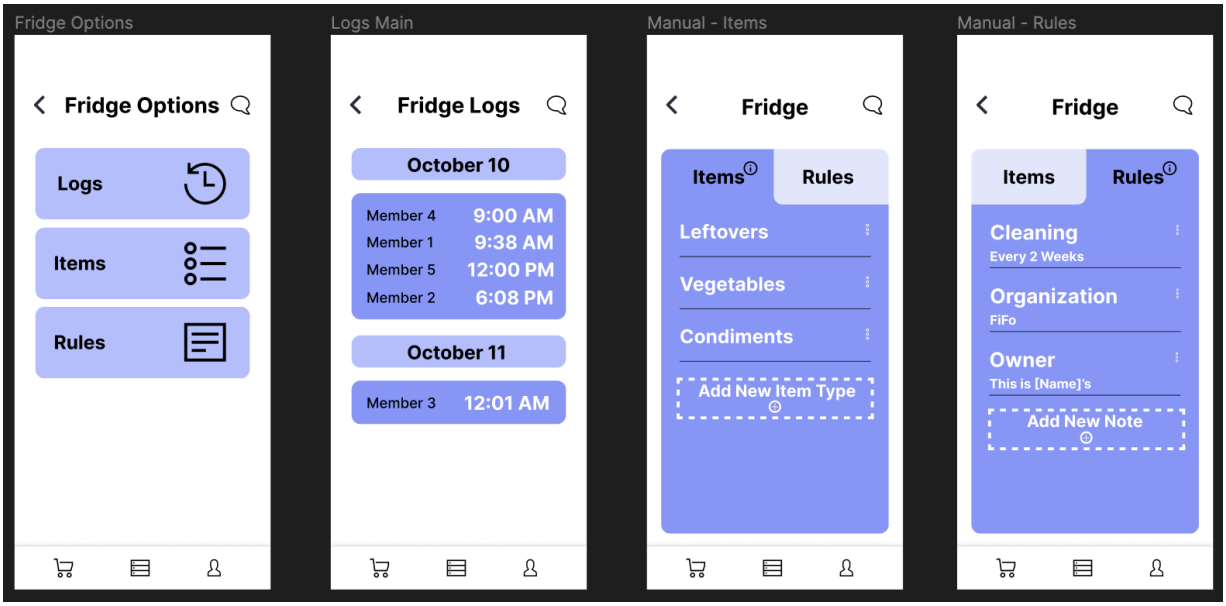


Figure 9. Logs and Rules

Usability Testing

The four tests that were used for this iteration are as seen below:

1. Scan a receipt and add the identified food items
2. Scan the receipt again, but edit the location of Fresh Milk from “Uncategorized/Others” storage area to “Pantry 1” storage area
3. Check the history log for the “Fridge” storage area
4. Check the rules set for the “Fridge” storage area

The results were once again listed on Google Sheets, and 3 different people from different households were taken for this test.

USER 1							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
Scan a receipt and add the identified food items	i	1	35		15	didnt saw the cam properly	TRUE

Scan the receipt again, but edit the location of Fresh Milk from “Uncategorized/Others” storage area to “Pantry 1” storage area	i	1	40		10	clicked on pantry 1 instead of others	TRUE
Check the history log for the “Fridge” storage area		0	10				TRUE
Check the rules set for the “Fridge” storage area		0	8				TRUE

USER 2							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
Scan a receipt and add the identified food items	i	1	32			didn't see cam immediately (tapped cart first)	TRUE
Scan the receipt again, but edit the location of Fresh Milk from “Uncategorized/Others” storage area to “Pantry 1” storage area		0	31				TRUE
Check the history log for the “Fridge” storage area		0	8				TRUE
Check the rules set for the “Fridge” storage area		0	10				TRUE

USER 3							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
Scan a receipt and add the identified food items	ii	2	10			ok din, nagkaerror lang kase di alam navigation ng app on first use	TRUE
Scan the receipt again, but edit the location of Fresh Milk from “Uncategorized/Others” storage area to “Pantry 1” storage area			20			ok mabilis	TRUE
Check the history log for the “Fridge” storage area	i		22			Marami pinipindot di mo alam kung san hahanapin", setting button is weird, consolidated log is better than log per storage	TRUE

Check the rules set for the “Fridge” storage area	i		20			Per storage setting is hassle, Usually people prefer overall rather than per personalized, tldr; general setting then customizable when want to customize. Extra info: Rules seems like a confusing word kaya namisunderstood yung feature's intent. It's supposed to be an instruction manual and something for the app to detect and help in organizing the scanned receipt. Maybe guidelines, personalize, or setting is better word daw.	TRUE

The 4th prototype has a success rate of 100% across all tasks. This showcases the effectiveness of the implemented changes. Task completion times reveal a slight increase in average errors for the first task, which indicates a need to slightly improve the design for the next iteration of the prototype. Tasks 2, 3, and 4 has minimal to no errors, indicating that the user can perform those tasks easily. The efficient completion of complex actions such as editing item locations and navigating history logs and rules sets suggests an overall positive user experience.

Based on the comments, however, there are a few improvements that can be made. For task 1, issues with camera visibility and navigation were noted, this emphasizes the need for improved guidance—especially for first-time users—or a need for redesigning. On the other hand, task 2 revealed that a user misclicked on “Pantry 1” instead of the intended storage area, this also suggests a need for guidance for first-time users. Tasks 3 and 4 highlighted the need for streamlined changes within the application. One user advocated for a consolidated log rather than storage-specific logs, along with a more general setting rather than per-storage customization. This was also accompanied by a suggestion to rename and clarify the “Rules” feature for better comprehension.

These insights from the users signify the importance of having a user-friendly application in subsequent iterations.

Checkpoint 3.4 (Final iteration of prototype)

Brainstorming

During our final brainstorming session, our team leader efficiently delegated tasks, assigning specific responsibilities for articles, design, and usability testing of the prototype to different members of the group. The team decided on a thoughtful approach to enhance the user experience. The group collectively decided to turn the history logs feature from something accessible per individual storage area, to a feature that encompasses all storage areas; with better accessibility to this feature through its placement on the home screen and the main screen of the virtual food storage. The team also recognized the need for a more intuitive symbol that effectively communicates the concept of scanning a receipt to add food items. Furthermore, the team collectively decided to address the labeling of the “rules” section. Recognizing the importance of clear and informative descriptors, the group decided on replacing “rules” with a more intuitive label, ‘Guidelines’, that more accurately conveys the content of the feature. These adjustments aim to make the prototype more user-friendly and improve the overall user experience.

In addition to these changes the group placed an emphasis on navigation testing during this final iteration. By focusing on navigation, the group aimed to identify and address potential challenges users might face while interacting with the prototype, such as misleading icons, and confusing navigation due to numerous factors. The collaborative efforts of the team underscore our shared goal of delivering a seamless and user-friendly experience to our users.

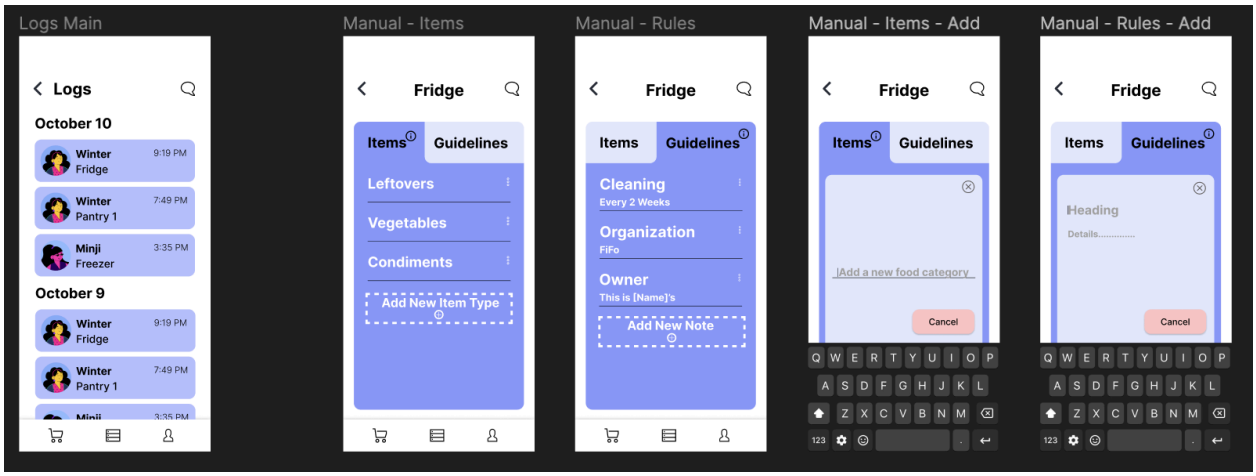


Figure 10. Redesigned history logs and items & guidelines screens

Usability Testing

The three tests that were used for this iteration are as seen below:

1. Check the history log which members accessed which storage area

2. Check the guidelines set for what should be done in the “Fridge” storage area
3. Add an item in the "Items" list for what should be added in the "Fridge" storage area

The results were once again listed on Google Sheets, and 2 different people from different households were taken for this test.

USER 1 (Nathan)							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
Check the history log which members accessed which storage area		0	3			Pretty fast	TRUE
Check the guidelines set for what should be done in the “Fridge” storage area	ii	2	27			Navigation requires proper exploring, it's deep and hidden	TRUE
Add an item in the "Items" list for what should be added in the "Fridge" storage area	ii	2	20			Navigation is still confusing	TRUE

USER 2 (Bien)							
Tasks	Errors	Error Count	Completion Time (seconds)	If failed, put max time here	Errors/Second	Comments	COMPLETION
Check the history log which members accessed which storage area		0	2				TRUE

Check the guidelines set for what should be done in the “Fridge” storage area		0	9			Settings icon is a bit misleading, can be replaced with the three-dot icon	TRUE
Add an item in the "Items" list for what should be added in the "Fridge" storage area		0	14			Not exactly for the items part, but the storage area's individual items could have a subtitle for which item type they fall into	TRUE

The final prototype has a success rate of 100% across all the tasks. This means that the changes that were applied based on the previous iteration's feedback and evaluation have been successful. For the task of checking the history log of members accessing a specific storage area, the absence of errors and a quick completion time of 2.5 seconds reflect a user-centered design. Tasks 2 and 3 show minimal errors, suggesting users can easily perform these tasks.

However, consistent feedback highlights confusion in navigation, indicating a need for overall design improvement and better guidance, particularly for new users. A specific user recommendation suggests replacing the settings icon with a three-dot icon for improved navigation, citing the current icon as misleading. Additionally, the suggestion to provide individual items within a storage area with descriptions specifying the type of food they categorize under the item name adds valuable input for enhancing the app's usability. Addressing these insights will contribute to refining the user experience in future iterations.

Final thoughts

How did you evaluate your prototypes?

During the initial stages, where the group was taking the parallel approach to creating low-fidelity prototypes from the problem statements, evaluation was conducted amongst the group members. Each member designed their own solution to a problem and gave feedback on the other members' solutions. This allowed the group to create faster and more focused solutions.

After the lo-fi prototypes were merged into one mid-fidelity prototype—which was created using the prototyping tool "Figma"—the group started to evaluate the prototypes through the use of usability testing every iteration, with 2-3 participants who followed the group's criteria for the app-based solution's target audience. The group provided these users with tasks to accomplish, and depending on the results of their tasks, it would then dictate whether or not there were any problems or errors that needed fixing. Whenever the tests for a feature would be a success, the feature was marked as usable, allowing the group to move on to testing and developing the next feature. However, if there were any issues that were to arise, the group would discuss the problems and come up with possible design changes to be implemented in the next iteration of the prototype. The feature was then placed in the list of tests again for the next iteration until no further errors arose.

What did you learn?

In the earlier stages of MCO2—specifically during the low-fidelity phase—we found out that going with parallel prototyping was the correct choice as we were able to take advantage of the fact that we all had different perspectives in the exploration and development of solutions for the problems that were observed from MCO1. This allowed the team to avoid any sort of excessive focus on one single solution or design, and allowed the team to be more focused on tackling individual problems with individual solutions rather than coming up with a solution that would have been more focused on compromising and making space for other solutions. Furthermore, this promoted creativity among the members as they were all free to come up with their own creative ideas on how to solve the problems that were observed from MCO1.

Thanks to the individual nature of parallel prototyping, we were also able to quickly point out, make suggestions, and modify early solutions made by the team members that included elements that would make the solution unfeasible to implement in real-life. An example of this is when a team member had to find a solution to a problem about how households have no efficient way of checking what food they had available at home without having to physically check all the food items for themselves. One of the first solutions for this was the use of a camera that had the ability to scan the food items in the storage spaces of households. This would have allowed the real-time remote tracking of what food items households had, along with the ability to check in real-time if a food item has been placed or removed from the storage area.

This solution was eventually scrapped since no such current consumer-grade technology exists to make it a feasible solution.

When we unanimously decided to move to mid-fi, The main learning we had come to understand was that there is no such thing as a “complete” feature or function. In the world of software development, while there are indeed objective criteria for when a feature can be considered complete and ready for deployment to the main codebase of a program, a feature is never fully “finished”. Even if it is considered usable and has no more bugs, the key to good software is continued change, in that design changes must continue to follow ever-changing design trends and user behavior not just to stay relevant but also to stay user-friendly.

The team also realized the importance of usability testing since we, as designers—from the moment we start working on the prototypes—-never fully understand the end user’s experience of using our program without actually getting feedback from those who are not familiar with our solution. Without usability testing, we would not be able to fully grasp the user interface and user experience-related issues that end users experience, such as the potential difficulty of a program’s navigation being too deep or being too broad, the thought process for why users would interact with certain elements of the prototype, or the overall flow users tend to follow when using a program. Usability testing gave us a lot of useful insights from the comments and suggestions made by the testers, which then allowed us to properly analyze the root problems our prototype had.

Was it a good solution after all?

Overall, the team can say that the final solution is a good solution, considering that the app-based solution was able to answer the problems that we observed from the needfinding study of MCO1. Moreover, there were considerably minimal errors performed and minor complaints given by the participants of our usability testing by the end of the mid-fidelity prototype phase. We were also able to quickly and efficiently apply all the necessary modifications and fixes based on the participants’ comments from the multiple usability tests we conducted. However, we believe that our solution can still be improved in the future as the team had a lot of ideas in mind on how we can further improve the user interface and user experience of the program that we were unable to implement properly due to lack of time and proper resources.